For low-vision users (not blind users). P1, iPhone on the cane (Sensor Logger, 100 Hz), recorded 2026-10-30: 17 sensor recordings, 11 videos lined up by sound, S10 hand-timed. Replaces v1 (2026-10-17). Grey numbers in brackets are v1 values. Forward-beam limits tuned on 2026-11-03 (x < 1.7 m, 0.3 m corridor). v3 (2026-11-06): the pod moves to a 130 cm cane. P1 recorded on a 120 cm cane, so every cane angle is mapped to the 130 cm cane at the same hand height, (L + grip) × sin α = constant (cane_analysis/cane_130cm.py), and the simulator is re-run on the mapped motion: β 19°, γ 73°, pause 52° / 49°. Arrows (→ 130 cm) mark the mapped values.

v4 (2026-11-07): the parts we really have. 2 × VL53L5CX (8×8 zones, 15 Hz, central 2×2 zones per beam), a Grove 3-axis accelerometer with no gyro, 2 × ELEGOO DC motors. Without a gyro the cane angle comes from the ground laser's floor plane (accelerometer in between), there is no sweep corridor and no walking speed. SIMULATED on P1's motion, accelerometer only, seed 7 (firmware/sim/results/validation_cane130_v0.81_lift0_beta19_accelonly_l5cx88.md): drops 18/20 (constant contact 10/10, two-point 8/10; median 0.44 m before the tip), flat-floor false alarms 0.27/min (two-point 0.54/min), clutter 9.1/min vs naive 20.8/min (walls and people beside the path alert without a corridor), head-level bar 20/20 (median 2.06 m before it), side of a head obstacle unknown (always "ahead"), median edge-to-motor 430 ms, calibration within ±3 cm (hold the cane still 3 s). The gyro numbers below are for the plan parts. The judging demo can also run on the demo build (simulated sensor data, real logic and motors: DEMO tab, section 10).

Design focus: low vision. Users see large obstacles and open space but misjudge depth: steps, curbs and platform edges at close range. So the pod (1) gives the earliest drop-off warning it can at their walking speed (0.8 m/s), (2) stays quiet once the user has found an edge and is looking at it, (3) keeps false alarms rare, because a user who can partly see the floor stops trusting a pod that buzzes for nothing, and (4) keeps head-level sensitivity for thin, low-contrast overhangs (branches, bars), which low vision misses.

Setup

Item	Value
Cane / pod	**130 cm cane (P1 recorded on 120 cm, hand 1.05 m from the tip). Pod 0.72 m from the tip = 58 cm below the top (slides 43–59 cm); hand about 15 cm below the top, so grip-to-pod 0.43 m and tip-to-hand 1.15 m**
Curb / stairs	curb 15 cm; stairs 13 steps × 14 cm
Two-point tip lift	8 cm by tape; 8.6 cm median from the cane angle (section 1)
Sync	3 taps at the start and the end of every recording. Videos (S06–S09, S11–S13, S15) matched to the sensors by sound: 10–31 cane impacts per video land on an audio click and

	all 3 start taps match; offsets −0.49 to +2.06 s. All 6 stair-top edges (S11, S12) are confirmed in the video within about 0.5 s of the sensor time.
Recorded	S1 natural + high, S2, S3, S4 (100 s), S5 (115 s), S6 step-down, S8 step-up ×2 (constant contact; two-point in the file named “stepdpwn_tap”), S9 left + right, S10 (2 timed runs), S11 ×3, S12 ×3, S13 ×3 descents, S15. Not recorded: S7, S14.

1. Measured

Item	**Value**
Data quality	100 Hz, gravity OK in all 17, gyro bias 0.01°/s
Static angle vs inclinometer	43.53° → phone 45.80° (+2.27) 59.45° → 61.0° (+1.53): phone mounted ~1.9° tilted (lying flat it reads 3.8°). Spread 0.7°
Walking cane angle α (S4 + S5)	shaft median 41.1°, 36.5–45.1° (phone 43.0°, 38.4–47.0°) → 130 cm: 36.9°, 32.9–40.3°
Highest α while walking	51.6° → 130 cm: 45.7°
α when the edge is found	35–46°, mean 42° (phone) → 130 cm: shaft 30–39°
α after the edge (peak)	curb 60–72°; stair top 62–67°; stair descent 65–83° (p5–p95), peaks 83–86° → 130 cm: curb 52–60°, stair top 54–57°, stairs 56–65°
Edge found → α > 57°	0.35–0.96 s (mean 0.65)
Sweep rate / half-width (p95)	0.45–0.49 Hz / 22.5–25.6°
Frames a 30 cm drop stays in view, worst (30 Hz)	S4 3.9 / S5 4.3 · approaches 3.6–4.0 → 3-frame fits. 130 cm: the beam spot is 8% farther from the hand → 3.3–4.0, 3 frames still fit
Walking speed (S10, 12.16 m)	0.78 and 0.84 m/s (mean 0.81)
Two-point tap rate	55 taps/min
Two-point tip lift from the cane angle	median 8.6 cm, p90 14 cm (S5, 102 taps); curb 10.1 cm, stair top 11.6 cm. Tapping lifts the tip mostly by pivoting the cane in the hand.

Stairs	13 × 14 cm; 13 tip impacts found in every descent; 1.39–1.55 s per step
α error, basic filter (K = 0.98)	p95 3.8° (sweep) – 6.6° (tap)
α error, firmware filter (τ = 2 s)	p95 2.2° (sweep) – 3.4° (tap); centripetal correction does not help overall (3.4° → 4.4°), so it stays off

2. Ground beam

Parameter	**Value**
Design α (coverage)	**37° (33–40°) on the 130 cm cane [120 cm: 41° (37–45°)]**
Tip-to-pod distance L	0.72 m
Mount angle β	**19° above the shaft axis (spares 18°, 20°); 18.8° exact for 0.75 m at 36.9°, 18.8–19.1° wherever the hand sits [120 cm: 21°]**
Pod height	0.432 m
Look-ahead past the tip	0.76 m (0.65–0.98 m over the α range)
Max look-ahead allowed by α error	about 1.17 m (7 cm budget at 3.4°)
ToF reading: flat / 14 cm step / 15 cm curb	1.41 m / 1.86 m / 1.90 m
Range needed	1.28–2.88 m (incl. a 30 cm drop at the lowest angle) → long mode
ROI / rate	about 15° (4×4 SPADs) / 30 Hz
Confirmation	3 frames (100 ms)
Δh sensitivity	1.33 cm per 1° α · 0.31 cm per 1 cm ToF
Alert	$\Delta h = L \cdot \sin\alpha - d \cdot \sin(\alpha - \beta)$ + tip lift < −8 cm

3. Forward (head) beam

Parameter	**Value**
Mount angle γ	73° above the shaft axis (36° above horizontal) [120 cm: 77°]

Elevation over the α range	33–40°
1.5 m overhang detected at	1.32–1.61 m ahead of the pod (1.46 m typical)
Head height (1.8 m) reached at	1.67–2.08 m ahead
Max range / ROI / rate	about 2.6 m (long mode) / 15–20° / 30 Hz, 2 frames
Alert	0.9 m < H < head + 0.10 m, x < 1.7 m (a 1.5 m overhang is met up to 1.61 m ahead), within 0.3 m of the walking line

4. Alert rules

Rule	**Value**
Drop threshold	Δh < −8 cm (curb 15 cm and stair 14 cm read −14 to −15 cm). No floor return at all = deep drop (platform edge) after 5 frames, but only when the empty run starts after at least 3 good floor readings on a floor that normally answers (poor above 30% empty frames over ~1 s, normal again below 15%: dark or shiny floor, sunlight); the call is kept for the whole run, so a real edge keeps alerting. Simulated with 30% empty frames: flat-floor false alarms 6.2 → 1.9/min, clutter 5.4 → 1.7/min, off-path box 7 → 1, drops still 20/20 (120 cm run)
Confirmation	ground 3 frames at 30 Hz (fewer false alarms, same detection); forward 2 frames (3 frames lose thin overhangs: 7 → 3 of 20 in simulation)
Two-point tip lift	compensated: lift = (L + grip) × (sin α at the last tap − sin α now), added back to Δh
Near-vertical pause	off above 52°, on below 49° (130 cm cane): walking never passes 45.7° after an edge the cane goes to 52–65° [120 cm: 60° / 55°, walking ≤ 51.6°, after an edge 60–86°], so the pod goes quiet once the user has found the edge
IMU filter	gravity-vector complementary filter, τ = 2 s, no centripetal correction, fast start for 2 s
Warning (0.76 m look-ahead, 100 ms	0.88 s at 0.78 m/s, 0.81 s at 0.84 m/s, 0.66 s

confirmation)	at 1 m/s

4b. Technique and walking speed (firmware, from the IMU)

The pod now recognises how the cane is used and estimates walking speed from the sweep arcs or taps. Checked by running the firmware on every real recording (timed speed in S10: 0.78–0.84 m/s):

Recording	Expected	Detected	Speed (firmware)
S04 walk	sweep	sweep 95%	0.84 m/s
S05 walk	tap	tap 94%	0.81 m/s
S09 left / right	sweep	sweep 91–100%	0.86 / 0.78 m/s
S11, S15 stair top	sweep	sweep 96–100%	0.75–0.78 m/s
S03 standing sweep	sweep, not walking	sweep 94%, walking 6%	–
S01, S02 still	still	still 100%	–

Rule	Value
Technique	sweep (yaw-rate RMS > 20°/s), tap (3 rhythmic impacts 0.5–1.8 s apart on sweep turning points), hold (walking without sweeping: diagonal cane), still; switches after 1 s
Walking	step bounce > 0.45 m/s² (standing sweep ≤ 0.41, walking ≥ 0.47)
Drop threshold per technique	sweep / hold / still 8 cm; tap 9 cm, plus tip-lift compensation
Speed	arcs or taps per second × 0.83 m per arc (sweep) or 0.89 m per tap (P1); one measured walk recalibrates it per user (serial: walk start, then walk 12.16)
Urgent drop	if look-ahead ÷ speed < 0.6 s: faster, stronger drop pattern (P1 at 0.81 m/s: 0.9 s, normal pattern; urgent above about 1.27 m/s)
Slow-down cue	speed > 1.3 m/s for 5 s: 2 soft pulses, at most once a minute
Session log	time per technique, distance, mean speed, alerts per walk (serial: stat) → real numbers for the poster

5. Simulated validation

Real P1 motion (S4 constant contact, S5 two-point with its real tip lift), virtual course. Re-run 2026-11-06 for the 130 cm cane: P1's IMU samples are tilted to the mapped 130 cm angle before the firmware sees them. Speed 0.81 m/s, 14 cm drop, beams 19° / 73°, pause 52° / 49°.

Test	Simulated result	Target
Drop-off (14 cm), 20 approaches	19/20 caught (constant 10/10, two-point 9/10); median 0.41 m before the tip, 3/20 at ≥ 0.5 m	≥ 18/20, ≥ 0.5 m
False alarms, empty flat floor	smart 1.65/min vs naive 14.8/min	≤ 1/min
False alarms, clutter course (walls, boxes, people beside the path, sign above head)	smart 0.58/min vs naive 21.4/min (37× fewer)	≤ 1/min, ≥ 5× fewer
Head-level 5 cm PVC bar at 1.5 m	10/20 (first alert 2.0 m before it)	≥ 18/20
Head-level 30 cm sign beside the path	14/20 caught, side correct 64%	side ≥ 85%
Off-path floor box	0 alerts in 10 passes	0
Latency, edge in beam → motor	median 60 ms	≤ 100 ms
Calibration at 3 pod positions	L within 1.1 cm; flat Δh within 1 cm	± 3 cm

Option compared (same course)	Drops caught	Median before tip	Flat-floor false alarms	Clutter false alarms
Chosen, 130 cm cane: β 19°, γ 73°, N = 3, tip lift compensated, tap threshold 9 cm, forward x < 1.7 m, corridor 0.3 m, pause 52° / 49°	19/20	0.41 m	1.65/min	0.58/min
130 cm cane, β 20° (spare mount, 0.85 m look-ahead)	19/20	0.48 m	2.19/min	1.39/min
120 cm cane (v2): β 21°, γ 77°, pause 60° / 55°	20/20	0.47 m	1.3/min	1.9/min
120 cm cane, β 19° (0.62 m look-ahead)	20/20	0.37 m	1.3/min	2.9/min

120 cm cane, β 19°, no tip-lift fix	20/20	0.37 m	18.6/min	20.6/min

130 cm cane: β 19° beats the 20° spare on false alarms (clutter 0.58 vs 1.39/min) for 0.07 m less margin, so 19° stays. By technique, constant contact: flat floor 0.6/min, clutter 0.6/min; two-point tapping: clutter 0.5/min but flat floor 2.7/min, the remaining gap (120 cm run: 2.7 and 3.2/min). Clutter alarms come mostly from the forward beam (people standing near the path). Narrowing the walking-line corridor from 0.4 to 0.3 m and extending the forward limit from 1.5 to 1.7 m (2026-11-03) cut them from 3.4 to 1.9/min and catch slightly more overhangs (bar 7 → 8/20, sign 13 → 14/20), but left/right is then only 57% correct (64% on the 130 cm cane): expect Plan B (rhythm only) unless the haptic test says otherwise. The 120 cm β 19° rows use the earlier limits (1.5 m, 0.4 m). The thin PVC bar fills too little of the forward beam; use a 20–30 cm board for the demo, or a wider forward ROI. If the hand also lifts the cane (not only pivots), compensation misses that part: an extra 3 cm of arm lift gives 4/min on a flat floor, so keep taps low in the demo.